TRANSFINITTE ’26 · DARE TO DREAM

SAATHI

A chair that warns of worsening heart failure, days before the hospital.

Sit, press green: a check designed for under 2 minutes a day. The family gets a text when readings stay above that person’s normal.

Team Wildcat

Heart failure worsens quietly at home

WEIGHT

Gain began at least a week before admission

Chaudhry et al., Circulation 2007. Some admissions came with no gain at all.

FLUID

Chest impedance fell about 2 weeks early

Yu et al., Circulation 2005, with implanted devices.

VITAL SIGNS

A chest patch caught 76–88% of events

Stehlik et al., Circ Heart Fail 2020: about 6.5 days early, but worn all day.

The signs show up days early. Someone has to measure them every day, in an Indian home, with almost no effort.

Figures as summarised in the problem statement’s evidence table

Why a chair

Already used every day

The check fits a habit the patient already has. Nothing to wear, charge or remember.

The stillest posture

Sitting quietly gives the cleanest heartbeat signals from the seat and fingertip.

One device, every sensor

Weight, foot impedance, heart rate, oxygen, breathing and a voice check in one place.

Useful all day

It is furniture first, so it stays in the home and in use for months.

Seven biomarkers, each chosen from the evidence

Biomarker · sensor	Weight	Why we chose it	Research basis
Body weight 4 load cells in the legs	1.0	Retained fluid shows as weight gain about a week before many admissions; also the +2 kg in 3 days rule.	Chaudhry et al., Circulation 2007
Foot bioimpedance 50 kHz, 4 foot plates	1.0	Tissue impedance falls as fluid builds, about 2 weeks before admission; 4 electrodes cancel skin contact.	Yu et al., Circulation 2005 · Kyle et al., Clin Nutr 2004
Resting heart rate seat BCG + finger PPG	0.7	The heart beats faster at rest as it compensates; seat and finger sensors confirm each beat.	Stehlik et al. (LINK-HF), Circ HF 2020 · Inan et al., Circ HF 2018
Breathing rate finger PPG, 3 methods	0.7	Breathing quickens as fluid reaches the lungs; three pulse-wave methods fused into one rate.	Stehlik et al. 2020 · Karlen et al., IEEE TBME 2013
Blood oxygen (SpO_2) finger red / infrared	0.5	Oxygen falls when fluid limits gas exchange; scored as a drop from the person's own normal.	Chan et al., Respir Med 2013
3 symptom questions voice + 2 buttons	0.7 each yes	Breathlessness, more pillows to sleep, swollen feet: the congestion signs patients notice first.	McDonagh et al., ESC HF Guidelines, Eur Heart J 2021
PTT · chair use BCG→PPG timing · sit-downs	0.25	Pulse transit time tracks blood-vessel changes; fewer sit-downs mean less daily activity.	Mukkamala et al., IEEE TBME 2015 · Stehlik et al. 2020

Weights fixed before testing by strength of evidence. Each signal scores only beyond the person's own normal (median ± MAD, Leys et al. 2013), so no single signal decides alone.

It compares each person only with themselves

Learn their normal

First 14 days set each signal's usual range. A new medicine dose becomes the new normal.

→

Score each day

Points only for the part beyond their normal. Weight and foot impedance count most; PTT least.

→

Wait for 2 days

One salty festival meal is not an alert. Two high days in a row are.

→

Alert with a reason

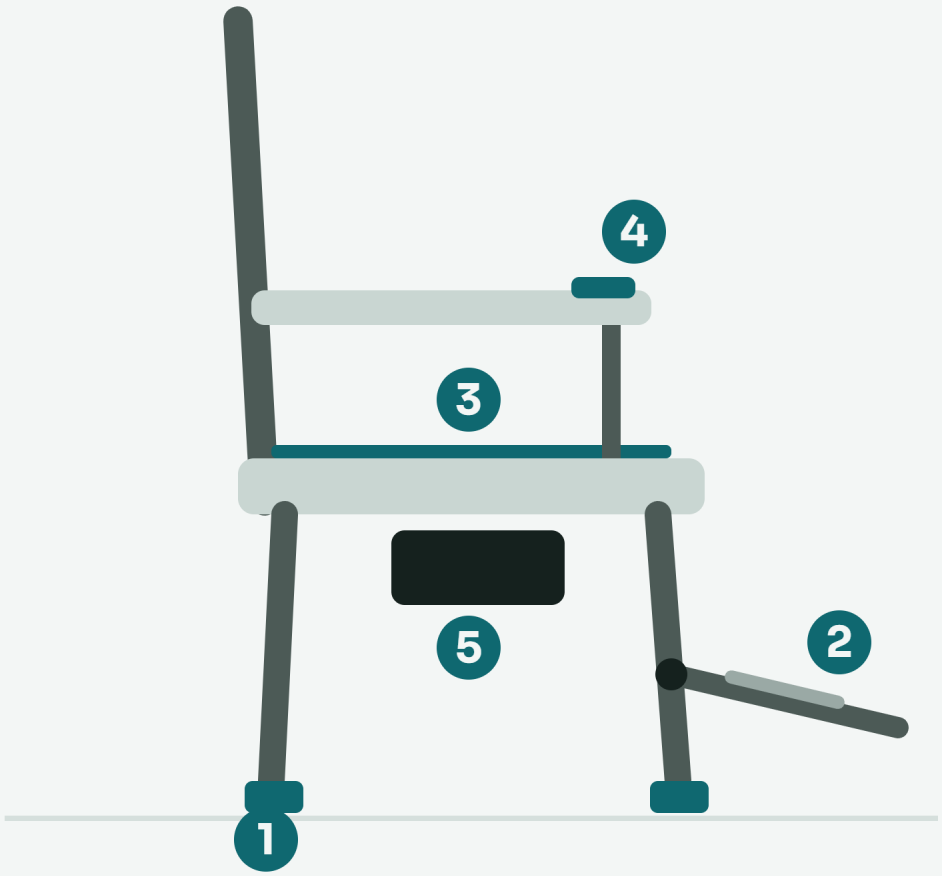
Danger rules skip the wait: +2 kg in 3 days, SpO_2 4 points below their normal twice, or checks stopping after a high day.

ALERT SMS TO THE FAMILY · PATIENT P006 · DAY 29

Readings above their normal 2 days running. Reported more foot swelling; Oxygen (SpO2) −2.1 points vs usual; Foot impedance −3% vs usual; Weight +0.5 kg vs usual. Please call the doctor today.

Sent 5 days before this synthetic patient's admission.

Every sensor lives inside the chair



1 Legs

A load cell in each foot weighs the whole body and notices when someone sits.

2 Fold-out footrest

Four steel plates (toes, heels) measure foot impedance. Hinged to the frame, so leg weight counts.

3 Seat

A PVDF film under the cushion feels each heartbeat (BCG).

4 Armrests

Right: finger sensor for pulse, oxygen, breathing. Left: green and red answer buttons, SOS under a flip cover.

5 Under the seat

Sealed box: ESP32, battery, SIM module, speaker. Works through power cuts.

Designed for under 2 minutes: 110–116 s

PATIENT · 12 s

Sit and press green

Bare feet on the plates, one finger in the groove. Any time of day.

→

PATIENT · 90 s

Sit still, answer 3 questions

Spoken in Tamil, Hindi or English during the 90 s. Left hand: green yes, red no.

→

PATIENT · 10 s

Hear the result

Normal, watch, alert, or “please repeat”.
Never silent.

→

FAMILY

Gets a text

Daily summary by SMS; an alert says why it fired.

→

DOCTOR

Sees trends

Each signal against the patient’s own normal range.

Why they keep using it for months

It is the chair they already sit in.

Forgot? It reminds them when they sit. Two missed days text the family; after a high reading, that text is an alert.

The SOS button makes it worth keeping even on good days.

80% caught on a locked test, scored once

80%

events caught

103 of 129 · 95% CI 72–86%

95.8%

normal days left alone

specificity

40%

of alerts were real

precision · 0.77 false alerts per 30 normal days

3 days

median warning

caught events · 50% of all warned ≥ 3 days

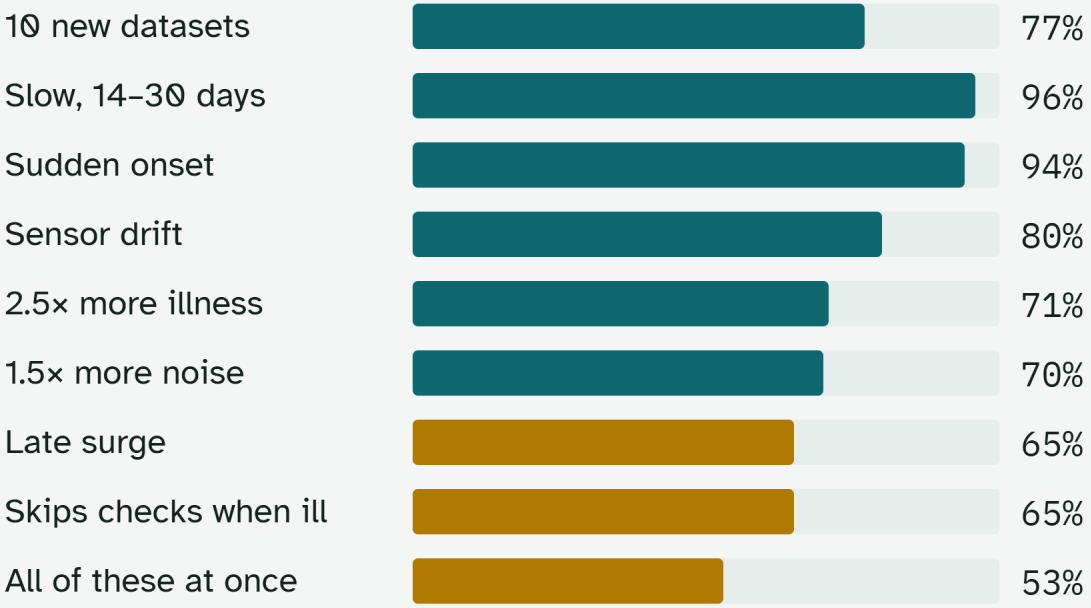
Locked test	Alert	No alert
Worsening event	103	26 missed
Normal day	276	6,249
Alert episodes	113 real	167 false

Protocol, seed and file hashes recorded **before** 200 new patients were generated; scored once; a re-run is refused (a reproducibility safeguard, not an independent audit). The usual **weight-only rule** catches **23%**.

Locked synthetic set: 200 patients, 10,151 patient-days. Normal days = after the 14-day learning period, outside worsening; per 30 recorded days: 0.49. By type: congestive 97%, no weight gain 72%, subtle 36%. Synthetic, not clinical evidence.

We tried to break it

Events caught when we change the patients



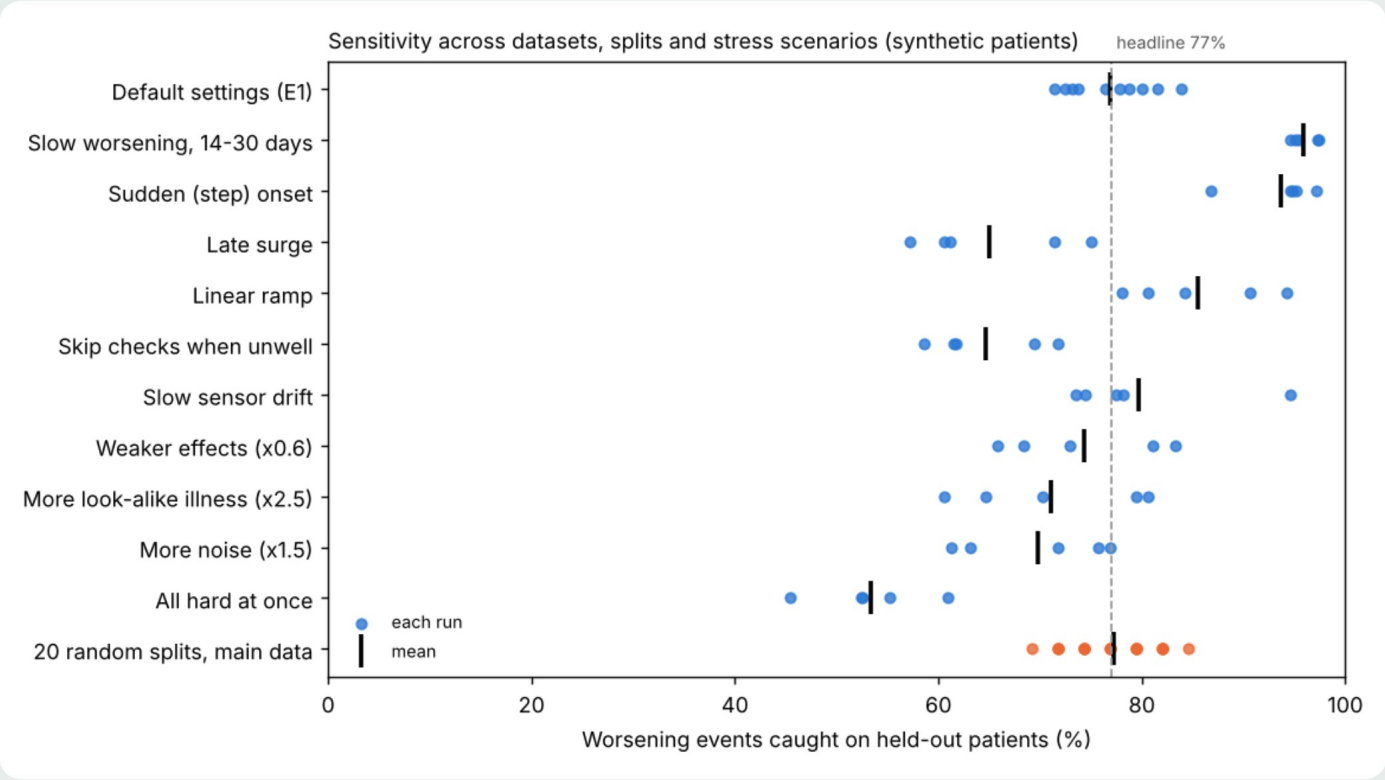
Full protocol repeated each time on new synthetic patients (5–10 datasets per row).

What each signal adds (development test set)

Version	Caught	False / 30 normal days
Full SAATHI	74%	0.83
Without breathing rate	72%	0.67
Without the 3 questions	69%	0.64
Lite: no foot impedance, PTT, chair use	77%	0.66
Weight-only rule	23%	0.11

Broken sensors: 14 faults injected into raw waveforms; a missing finger gave a wrong SpO_2 in 95% of checks before the quality gates, 0% after. **Real recordings:** beat detector within 0.03 bpm of expert ECG labels.

Stress-tested against what real life throws at it



HOW WORSENING UNFOLDS

Slow over 14–30 days, sudden step, late surge, steady ramp

HOW PATIENTS BEHAVE

Skipping the check on the days they feel unwell

HOW SENSORS AGE

Slow drift (load-cell creep, electrodes), 1.5× more noise

HOW HARD THE SIGNS ARE

Weaker changes (0.6×), 2.5× more look-alike illness: fevers, festival meals

EVERYTHING AT ONCE · 20 SPLITS

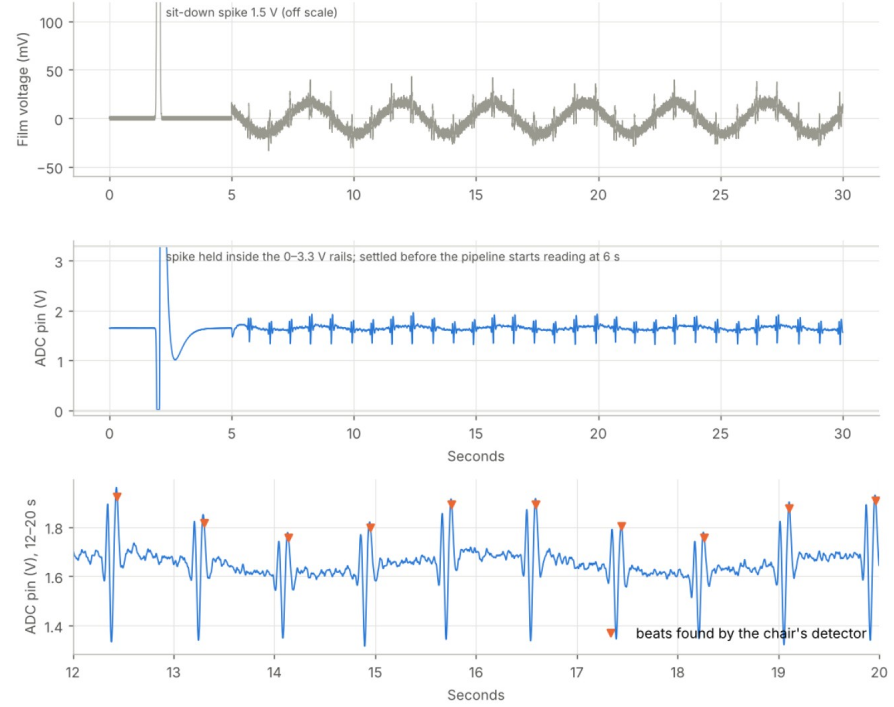
All the above together; 20 random train/test splits of the main data

Each dot is a full run on new synthetic patients; bars are the mean; dashed line = 77% headline across 10 fresh datasets.

₹4,661 for the whole chair

Block	₹
Chair (Nilkamal CHR2060)	1,040
Sensors and front-ends	1,430
Power: 2 cells, charger, boost	511
ESP32 controller	499
SIM800L for SMS	364
Voice: amplifier and speaker	42
Buttons, wiring, footrest, mounts	775
Total, under ₹5,000	4,661

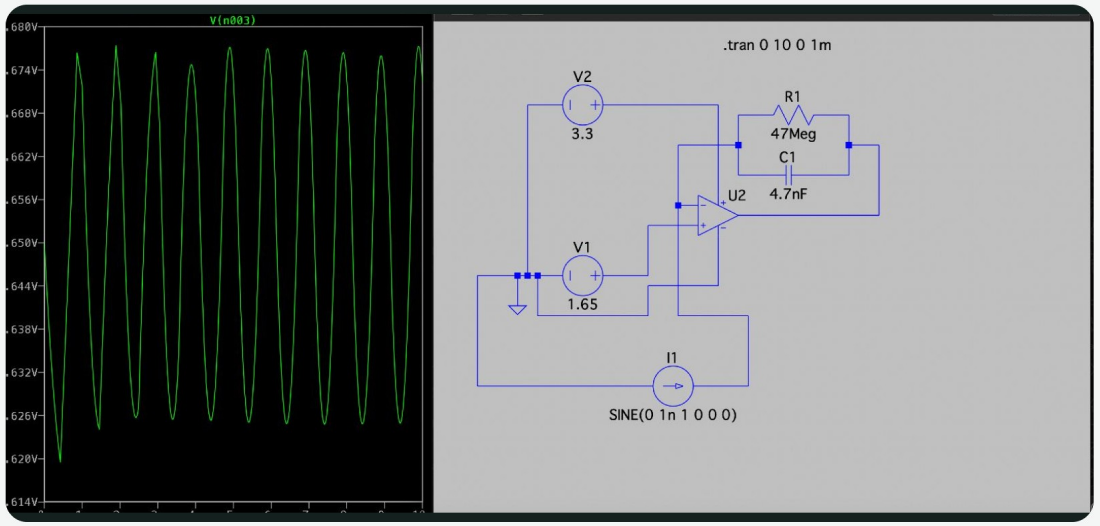
₹3,504 from online e-commerce websites (checked 24–25 Oct 2026); ₹1,157 estimated. **Conservative** (out-of-stock swaps + 10% on estimates): ₹4,912. **Lite** (no foot impedance): ₹4,259.



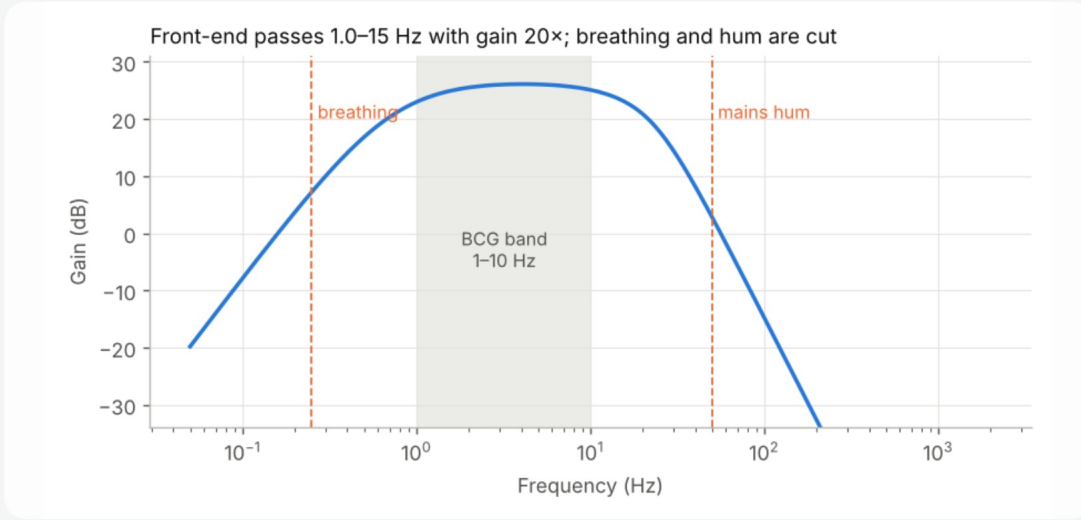
Seat heartbeat circuit simulated in LTspice: heart rate from its output **72.1 bpm** vs **72.2** true; breathing cut 9×, mains hum 15×.

Battery through power cuts · offline storage · alerts queued, retried when network returns · no impedance test on mains or with a pacemaker

The seat heartbeat circuit, simulated in LTspice



Stage 1, charge amplifier. A 1 nA film signal gives ±27 mV, matching the hand calculation (47 MΩ ‖ 4.7 nF at 1 Hz = 27.5 mV).



Full front-end. Passes the heartbeat band (1.0–15 Hz, gain 20×); breathing cut 9×, mains hum cut 15×.

72.1 bpm

heart rate from the output vs 72.2 true

2–100 mV

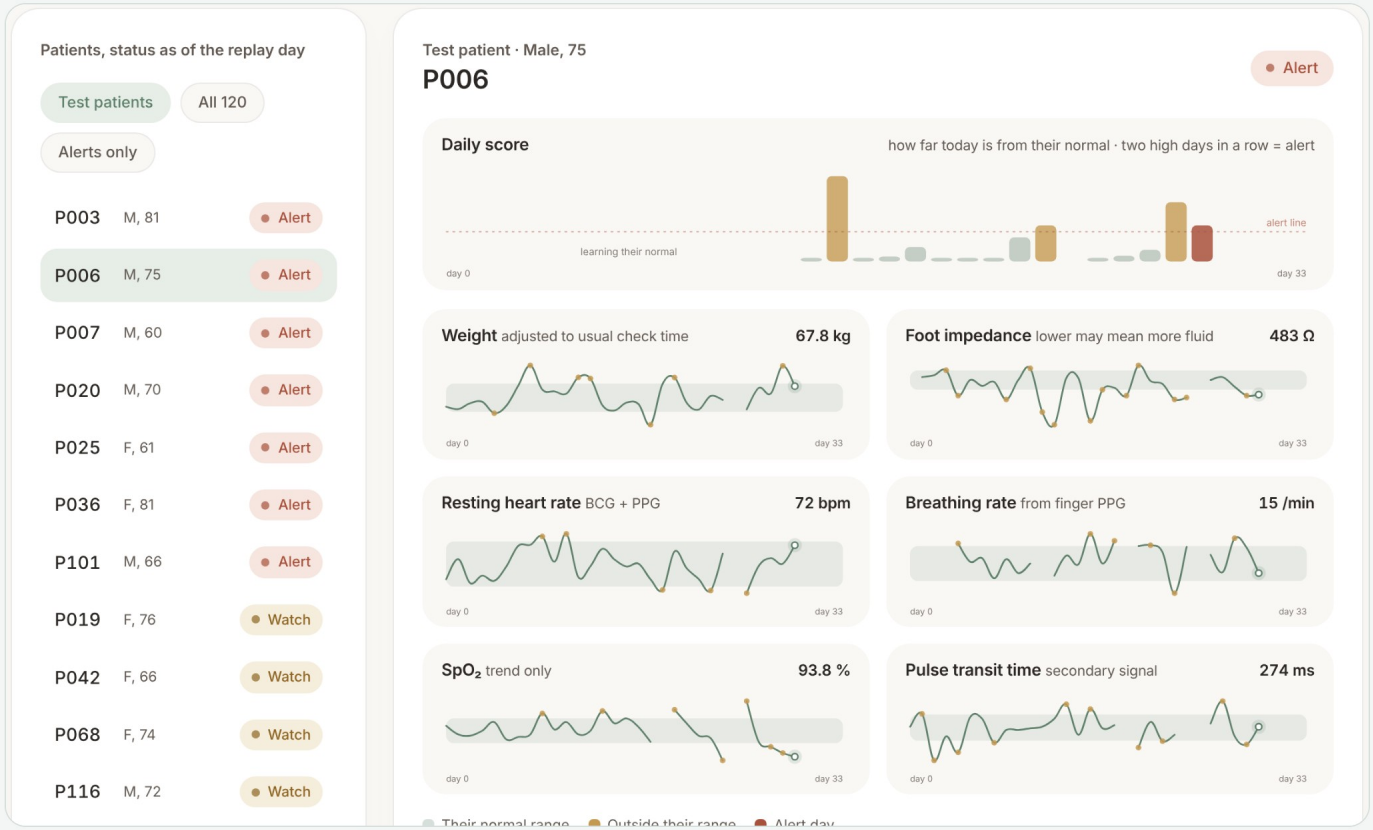
film signals all give the correct heart rate

0–3.3 V

sit-down spike held inside the ESP32's rails

PVDF film (TE LDT0-028K) → charge amplifier → band-pass → ESP32 ADC · MCP6004 op-amp, single 3.3 V supply

Demo: 28 simulated days through the real code



DOCTOR VIEW · ONE SIMULATED MONSOON

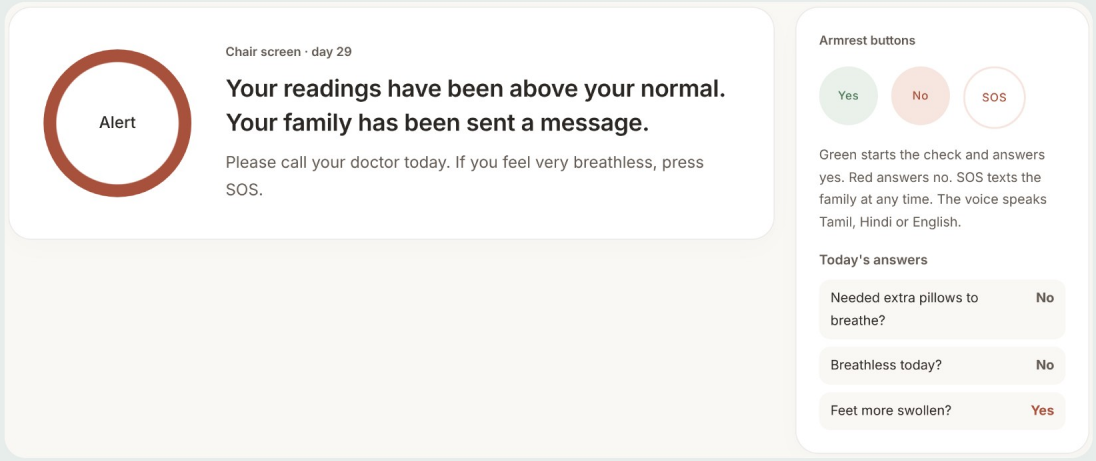
Day 15	Grandson sits: the chair asks who it is, no false alarm
Day 19	No finger: oxygen shown as unavailable, not normal
Day 21	Slow worsening begins
Day 24	Alert raised while the network is down; queued in the chair
Day 26	Signal returns: the family gets the SMS

> Day 28 **Admission: 4 days of warning**

python demo_scenario.py --step · real pipeline, detector and firmware SMS outbox · dashboard marked SIMULATION

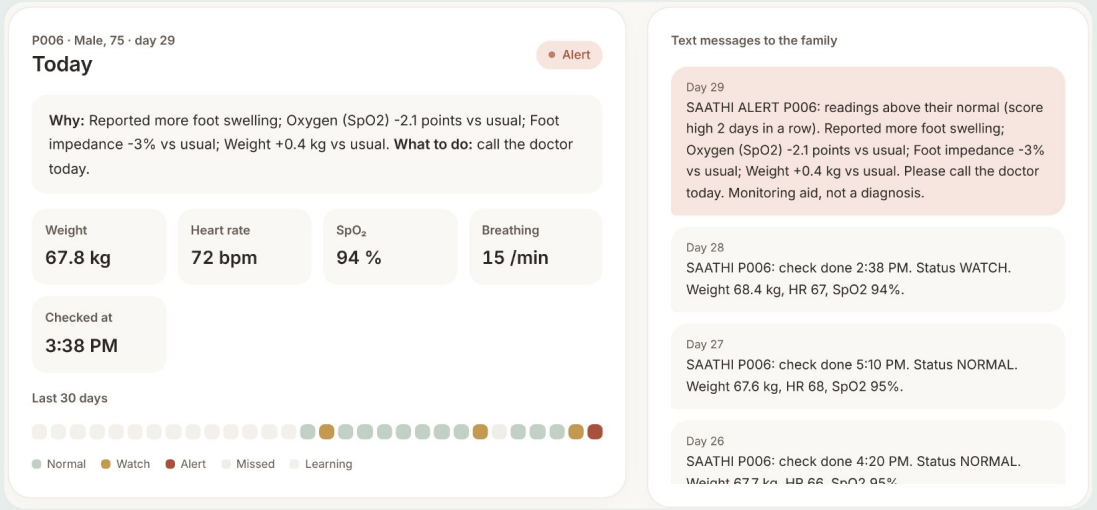
What the patient and the family see

PATIENT · THE CHAIR'S SCREEN



One status in plain words, what to do next, and today's three answers. Spoken aloud in Tamil, Hindi or English.

FAMILY · TODAY AND THE SMS LOG



Why it alerted, today's readings, 30 days at a glance, and every SMS the chair has sent.

Offline dashboard, simulated patient P006, day 29 · works without internet · marked SIMULATION

Every result is labelled: simulated, real signals, or not measured

Tested in software

- Locked test + stress tests on synthetic patients
- 14 sensor faults: no confident wrong number (1 known gap)
- Firmware = Python: every day, 9 safety cases, raw signals
- Regression test for every safety fix
- Beat detection on real public recordings

Simulated

- Both circuits in LTspice, incl. every part failing
- Seat circuit works for 2–100 mV signals
- About 20 days on battery (datasheets)
- 110–116 s per check (design target from real voice clips)

Not yet tested

- A physical chair
- Real seat-film signal and weight accuracy
- Use by older people
- Any real patient: needs ethics approval

Next milestone: build the weight and finger-sensor chain and run bench tests 1–2, then the BIDMC heart-failure ECG check, then, with ethics approval, a small pilot. **Simulation-first:** no physical chair was built.

SAATHI

Sit. Press green. See changes sooner.

Designed to support earlier awareness of changes. An ordinary chair, a check designed for under 2 minutes, ₹4,661 in parts. 80% of worsening caught on a locked synthetic test, with the reason in every alert.

Code, tests and setup: https://github.com/Shan-boop-cmd/Saathi_Chair · Monitoring aid, not a diagnosis · All patient data computer-generated